Coding & Solution

Date	29 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality and completeness of the OptiCrop: Smart Agricultural Production Optimization Engine. The project demonstrates the development of a machine learning-based crop recommendation system that analyzes soil and environmental parameters to predict suitable crops. The solution follows clean coding practices, modular design, and proper integration of machine learning models with a Flask web application.

Instructions:

1. Ensure the Python, Flask, HTML, and CSS code is properly structured, documented, and follows coding standards and best practices.
2. The solution should address the agricultural crop recommendation problem and implement key functionalities such as data preprocessing, model training, prediction generation, and web-based user interaction.
3. Include all source files, datasets, model files, configuration files, and setup instructions required to run the OptiCrop application.
4. Attach or provide the GitHub repository link containing the complete project source code and documentation.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mnvnarasimhulu4-cmd/AI-ML-OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Key Features Implemented	Crop recommendation prediction, data preprocessing, data visualization, K-Means clustering, Logistic Regression model, Flask web application, interactive prediction form, model deployment
Pending / Incomplete Features	Cloud deployment, advanced exception handling, additional machine learning algorithms, mobile responsiveness improvements
Setup / Run Instructions	Install Python and required libraries using pip install -r requirements.txt. Run app.py using Python and open the generated localhost URL in the browser to access the application.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project follows a modular architecture with separate components for data preprocessing, machine learning model development, and web application deployment. The system successfully predicts suitable crops based on soil and environmental conditions and demonstrates the practical implementation of AI and Machine Learning techniques in smart agriculture.